

DSA0501 – QUERY PROCESSING WITH DATA ANALYSIS

CO2 – AT1

NAME: Rakshitha V B

REGISTER NO: 192324004

QUESTION 1:

```
import sqlite3
```

```
conn = sqlite3.connect("college.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Student(
```

```
    Student_ID INTEGER PRIMARY KEY,
```

```
    Name TEXT,
```

```
    Department TEXT,
```

```
    Age INTEGER,
```

```
    CGPA REAL
```

```
)
```

```
""")
```

```
students = [
```

```
    (101, "Aarav", "CSE", 19, 8.9),
```

```
    (102, "Diya", "AI&DS", 20, 9.1),
```

```
    (103, "Rahul", "ECE", 19, 8.5),
```

```
    (104, "Sneha", "IT", 20, 9.0),
```

```
    (105, "Karan", "MECH", 21, 8.2)
```

```
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Student VALUES (?, ?, ?, ?, ?)", students)
```

```
conn.commit()

# Display records

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

print("Student Records")

for row in rows:

    print(row)

conn.close()
```

OUTPUT:

```
==
Student Records
(101, 'Aarav', 'CSE', 19, 8.9)
(102, 'Diya', 'AI&DS', 20, 9.1)
(103, 'Rahul', 'ECE', 19, 8.5)
(104, 'Sneha', 'IT', 20, 9.0)
(105, 'Karan', 'MECH', 21, 8.2)
>>>
```

QUESTION 2:

```
import sqlite3

conn = sqlite3.connect("library.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Book(

    Book_ID INTEGER PRIMARY KEY,

    Title TEXT,

    Author TEXT,
```

```

    Publisher TEXT,

    Price REAL

)

)

books = [

    (1, "Python Basics", "John", "ABC", 450),

    (2, "Java Programming", "James", "XYZ", 550),

    (3, "DBMS", "Navathe", "Pearson", 700),

    (4, "AI Fundamentals", "Russell", "McGraw", 800),

    (5, "Data Science", "Jake", "OReilly", 900)

]

cursor.executemany("INSERT OR IGNORE INTO Book VALUES (?, ?, ?, ?, ?)", books)

# Update price

cursor.execute("UPDATE Book SET Price=750 WHERE Book_ID=3")

conn.commit()

print("Updated Book Details")

cursor.execute("SELECT * FROM Book WHERE Book_ID=3")

print(cursor.fetchone())

conn.close()

OUTPUT:

```

```
>>> Updated Book Details
      (3, 'DBMS', 'Navathe', 'Pearson', 750.0)
```

QUESTION 3:

```
import sqlite3
```

```
conn = sqlite3.connect("company.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Department(
```

```
    Dept_ID INTEGER PRIMARY KEY,
```

```
    Dept_Name TEXT
```

```
)
```

```
""")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Employee(
```

```
    Emp_ID INTEGER PRIMARY KEY,
```

```
    Emp_Name TEXT,
```

```
    Salary REAL,
```

```
    Dept_ID INTEGER,
```

```
    FOREIGN KEY(Dept_ID) REFERENCES Department(Dept_ID)
```

```
)
```

```
""")
```

```
departments = [
```

```
    (1, "HR"),
```

```
    (2, "IT"),
```

```

        (3, "Finance")
    ]

    employees = [
        (101, "Rahul", 50000, 2),
        (102, "Sneha", 45000, 1),
        (103, "Amit", 60000, 3)
    ]

    cursor.executemany("INSERT OR IGNORE INTO Department VALUES (?,?)", departments)
    cursor.executemany("INSERT OR IGNORE INTO Employee VALUES (?, ?, ?, ?)", employees)

    conn.commit()

    cursor.execute("""
SELECT Employee.Emp_Name, Department.Dept_Name
FROM Employee
JOIN Department
ON Employee.Dept_ID=Department.Dept_ID
""")

    print("Employee - Department")
    for row in cursor.fetchall():
        print(row)

    conn.close()

```

OUTPUT:

```
Employee - Department
('Rahul', 'IT')
('Sneha', 'HR')
('Amit', 'Finance')
>>> |
```

QUESTION 4:

```
import sqlite3
```

```
conn = sqlite3.connect("hospital.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Patient(
```

```
    Patient_ID INTEGER PRIMARY KEY,
```

```
    Name TEXT NOT NULL,
```

```
    Age INTEGER,
```

```
    Contact TEXT,
```

```
    Disease TEXT
```

```
)
```

```
""")
```

```
patients = [
```

```
    (1, "Ravi", 35, "9876543210", "Fever"),
```

```
    (2, "Priya", 28, "9876500000", "Cold"),
```

```
    (3, "Anil", 40, "9999999999", "Diabetes"),
```

```
    (4, "Neha", 32, "8888888888", "Asthma")
```

```
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Patient VALUES (?, ?, ?, ?, ?)", patients)
```

```
# Update contact
```

```
cursor.execute("UPDATE Patient SET Contact='9000000000' WHERE Patient_ID=2")
```

```
# Delete patient
```

```
cursor.execute("DELETE FROM Patient WHERE Patient_ID=4")
```

```
conn.commit()
```

```
print("Remaining Patients")
```

```
cursor.execute("SELECT * FROM Patient")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:

```
Remaining Patients
(1, 'Ravi', 35, '9876543210', 'Fever')
(2, 'Priya', 28, '9000000000', 'Cold')
(3, 'Anil', 40, '9999999999', 'Diabetes')
>>>
```

QUESTION 5:

```
import sqlite3
```

```
conn = sqlite3.connect("course.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Student(
```

```
    Student_ID INTEGER PRIMARY KEY,
```

```
    Name TEXT
```

```
)  
""")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Course_Registration(  
    Reg_ID INTEGER PRIMARY KEY,  
    Student_ID INTEGER,  
    Course_Name TEXT,  
    FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID)  
)
```

```
)  
""")
```

```
students = [  
    (1, "Aarav"),  
    (2, "Diya"),  
    (3, "Rahul")  
]
```

```
registrations = [  
    (101, 1, "Python"),  
    (102, 2, "Machine Learning"),  
    (103, 3, "DBMS")  
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Student VALUES (?,?)", students)
```

```
cursor.executemany("INSERT OR IGNORE INTO Course_Registration VALUES (?,?,?)",  
registrations)
```

```
conn.commit()
```



```

cursor.execute("""
SELECT Student.Name, Course_Registration.Course_Name
FROM Student
JOIN Course_Registration
ON Student.Student_ID=Course_Registration.Student_ID
""")

print("Student Course Details")

for row in cursor.fetchall():

    print(row)

conn.close()

```

OUTPUT:

```

>>> | Student Course Details
      | ('Aarav', 'Python')
      | ('Diya', 'Machine Learning')
      | ('Rahul', 'DBMS')

```

QUESTION 6:

```

import sqlite3

conn = sqlite3.connect("bank.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Account(

```

```
Account_No INTEGER PRIMARY KEY,  
Customer_Name TEXT,  
Account_Type TEXT,  
Balance REAL CHECK(Balance>=0)  
)  
""")
```

```
accounts = [  
    (1001, "Rahul", "Savings", 15000),  
    (1002, "Priya", "Current", 25000),  
    (1003, "Anil", "Savings", 18000)  
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Account VALUES (?, ?, ?, ?)", accounts)
```

```
cursor.execute("UPDATE Account SET Balance=30000 WHERE Account_No=1002")
```

```
conn.commit()
```

```
print("Customer Records")
```

```
cursor.execute("SELECT * FROM Account")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:

```
Customer Records
(1001, 'Rahul', 'Savings', 15000.0)
(1002, 'Priya', 'Current', 30000.0)
(1003, 'Anil', 'Savings', 18000.0)
>>>|
```

QUESTION 7:

```
import sqlite3
```

```
conn = sqlite3.connect("inventory.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Product(
```

```
    Product_ID INTEGER PRIMARY KEY,
```

```
    Product_Name TEXT,
```

```
    Category TEXT,
```

```
    Quantity INTEGER,
```

```
    Unit_Price REAL
```

```
)
```

```
""")
```

```
products = [
```

```
    (1, "Laptop", "Electronics", 8, 55000),
```

```
    (2, "Mouse", "Electronics", 25, 500),
```

```
    (3, "Keyboard", "Electronics", 6, 900),
```

```
    (4, "Chair", "Furniture", 12, 3000)
```

```
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Product VALUES (?, ?, ?, ?, ?)", products)
```

```
cursor.execute("UPDATE Product SET Quantity=20 WHERE Quantity<10")
```

```
conn.commit()
```

```
print("Updated Inventory")
```

```
cursor.execute("SELECT * FROM Product")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:

```
Updated Inventory
(1, 'Laptop', 'Electronics', 20, 55000.0)
(2, 'Mouse', 'Electronics', 25, 500.0)
(3, 'Keyboard', 'Electronics', 20, 900.0)
(4, 'Chair', 'Furniture', 12, 3000.0)
>>>|
```

QUESTION 8:

```
import sqlite3
```

```
conn = sqlite3.connect("university.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Faculty(
```

```
    Faculty_ID INTEGER PRIMARY KEY,
```

```
    Faculty_Name TEXT
```

```
)  
""")
```

```
cursor.execute("""  
CREATE TABLE IF NOT EXISTS Course(  
    Course_ID INTEGER PRIMARY KEY,  
    Course_Name TEXT,  
    Faculty_ID INTEGER,  
    FOREIGN KEY(Faculty_ID) REFERENCES Faculty(Faculty_ID)  
)  
""")
```

```
faculty = [  
    (1, "Dr. Kumar"),  
    (2, "Dr. Priya"),  
    (3, "Dr. John")  
]
```

```
courses = [  
    (101, "Python", 1),  
    (102, "AI", 2),  
    (103, "DBMS", 3)  
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Faculty VALUES (?,?)", faculty)  
cursor.executemany("INSERT OR IGNORE INTO Course VALUES (?,?,?)", courses)  
  
conn.commit()
```

```
cursor.execute("""  
SELECT Faculty.Faculty_Name, Course.Course_Name  
FROM Faculty  
JOIN Course  
ON Faculty.Faculty_ID=Course.Faculty_ID  
""")
```

```
print("Faculty and Courses")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:

```
Faculty and Courses  
( 'Dr. Kumar', 'Python' )  
( 'Dr. Priya', 'AI' )  
( 'Dr. John', 'DBMS' )  
>>> |
```

QUESTION 9:

```
import sqlite3
```

```
conn = sqlite3.connect("vehicle.db")
```

```
cursor = conn.cursor()
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Vehicle(  
    Registration_No TEXT PRIMARY KEY,
```

```
    Owner_Name TEXT,
```

```
    Owner_Name TEXT,
```

```

        Vehicle_Type TEXT,

        Model TEXT

    )

    """)

vehicles = [

    ("TN01AB1234", "Rahul", "Car", "Hyundai"),

    ("TN02CD5678", "Priya", "Bike", "Honda"),

    ("TN03EF9999", "Anil", "Car", "Toyota")

]

cursor.executemany("INSERT OR IGNORE INTO Vehicle VALUES (?, ?, ?, ?)", vehicles)

conn.commit()

reg = "TN02CD5678"

cursor.execute("SELECT * FROM Vehicle WHERE Registration_No=?", (reg,))

print("Vehicle Details")

print(cursor.fetchone())

conn.close()

```

OUTPUT:

```

==
Vehicle Details
('TN02CD5678', 'Priya', 'Bike', 'Honda')
>>>

```

QUESTION 10:

```
import sqlite3

conn = sqlite3.connect("appointment.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Doctor(
    Doctor_ID INTEGER PRIMARY KEY,
    Doctor_Name TEXT,
    Specialization TEXT
)
""")

cursor.execute("""
CREATE TABLE IF NOT EXISTS Appointment(
    Appointment_ID INTEGER PRIMARY KEY,
    Patient_Name TEXT,
    Doctor_ID INTEGER,
    Appointment_Date TEXT,
    FOREIGN KEY(Doctor_ID) REFERENCES Doctor(Doctor_ID)
)
""")

doctors = [
    (1, "Dr. Kumar", "Cardiology"),
    (2, "Dr. Priya", "Neurology"),
    (3, "Dr. John", "Orthopedics")
]
```



```
appointments = [  
    (101, "Rahul", 1, "2026-07-30"),  
    (102, "Priya", 2, "2026-08-01"),  
    (103, "Anil", 3, "2026-08-02")  
]
```

```
cursor.executemany("INSERT OR IGNORE INTO Doctor VALUES (?, ?, ?)", doctors)
```

```
cursor.executemany("INSERT OR IGNORE INTO Appointment VALUES (?, ?, ?, ?)", appointments)
```

```
conn.commit()
```

```
cursor.execute("""  
SELECT Doctor.Doctor_Name,  
Appointment.Patient_Name,  
Appointment.Appointment_Date  
FROM Doctor  
JOIN Appointment  
ON Doctor.Doctor_ID=Appointment.Doctor_ID  
""")
```

```
print("Doctor Appointment Details")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
conn.close()
```

OUTPUT:

```
Doctor Appointment Details
('Dr. Kumar', 'Rahul', '2026-07-30')
('Dr. Priya', 'Priya', '2026-08-01')
('Dr. John', 'Anil', '2026-08-02')
```